

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5007		
Subject Title	MEMS (Microelectromechanical Systems) and Sensors		
Credit Value	3		
Level	5		
Pre-requisite / Co-requisite/ Exclusion	Nil		
Objectives	Study the recent progress on the design and fabrication of MEMS. Understanding the detection and actuation principles of MEMS. The applications of MEMS in sensing of motional, thermal, optical, chemical, and biological characteristics. Integration of MEMS with CMOS integrated circuits.		
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) provide in-depth information on the high demand for commercial and industrial applications of MEMS; (b) design suitable performance of MEMS with the specific performance; (c) fabricate simple MEMS on various platforms (i.e. fabrication on silicon or non-silicon based substrate); (d) utilize various designs of MEMS for different sensing capabilities; and (e) understand the possibility for the integration of MEMS with an electronics integrated circuit through the use of CMOS technique.		
Subject Synopsis/ Indicative Syllabus	MEMS overview: Introduction to MEMS and its applications, MEMS classification, industrial growth of MEMS, MEMS challenges and technology trends. Design and fabrication of MEMS: Knowledge of device physics includes beam theory, electrostatic and thermal actuation, capacitive and piezoresistive effects for the design of MEMS. The fabrication process of MEMS such as silicon etching, wafer bonding and etch stops, specific MEMS fabrication technologies, bulk micromachining processes, surface micromachining processes, SOI deep RIE process, non-silicon MEMS micromachining processes and more. Applications of MEMS as sensors: The use of MEMS to perform mechanical (motional) sensing, optical sensing, thermal sensing, magnetic sensing, as well as chemical and biological sensing. CMOS-based MEMS technologies: The concepts to integrate MEMs with CMOS circuitry on a single chip. The advantages of miniaturization and performance improvement of sensors.		
Teaching/Learning Methodology	Lecture: The fundamentals of MEMS and its applications as sensors will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance. Tutorial: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions.		
Assessment Methods in Alignment with	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)

Intended Learning Outcomes			a	b	c	d	e
	1. Continuous Assessments (Assignments and tests)	40%	✓	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓	✓
	Total	100 %					
	Continuous assessments consist of assignments and mid-term test. These will assess the students' understanding of basic concepts and principles in MEMS technologies for sensing purposes. The examination will be conducted to make a comprehensive assessment of students' intended learning outcomes as stated above.						
Student Study Effort Expected	Class contact:						
	▪ Lecture					33 Hrs.	
	▪ Tutorial					6 Hrs	
	Other student study efforts:						
	▪ Reading and self-study					81 Hrs.	
	The total student study effort					120 Hrs.	
Reading List and References	An Introduction to MEMS, Prime Faraday Technology Watch – January 2002, ISBN 1-84402-020-7 https://www.lboro.ac.uk/microsites/mechman/research/ipm-ktn/pdf/Technology_review/an-introduction-to-mems.pdf). Nadim Maluf and Kirt Williams, An Introduction to Microelectromechanical Systems Engineering, 2nd Edition, Artech House Print, 2004, https://www.zuj.edu.jo/download/an-introduction-to-mems-engineering-nadim-maluf-and-kirt-williams-pdf/) H. Qu, and H. Xie, CMOS MEMS Fabrication Technologies. In: Bhushan B. (eds) Encyclopedia of Nanotechnology. Springer, Dordrecht, 2012. https://link.springer.com/referenceworkentry/10.1007%2F978-90-481-9751-4_310) S Nihtianov, A. Luque, Smart Sensors and MEMS, 2nd Edition, Woodhead Publishing, 2018. Goutam Koley and Ifat Jahangir (Eds.), MEMS/NEMS Sensors – Fabrication and Application, MDPI, 2019 (https://www.mdpi.com/books/pdfview/book/1827). Zhuoqing Yang (Ed), Advanced MEMS/NEMS - Fabrication and Sensors, Springer, 2022 (https://link.springer.com/book/10.1007/978-3-030-79749-2).						